Brushless DC Motor Design Project

Design, CAD Modelling, 3D Printing, Winding, Assembly, and Testing of a Low Power BLDC Motor

Vivek Aruja

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Abstract

This project covers the design, modelling, fabrication, winding, assembly, and testing of a small brushless DC motor. The objective was to design a radial flux, three phase, six slot/eight pole outrunner BLDC motor that satisfied the course size and electrical constraints and demonstrated the full motor design workflow from calculations to physical testing. The motor targets a rated torque of $0.075\,\mathrm{N} * \mathrm{m}$, a rated speed of 2000 RPM, a rated voltage of $12\,\mathrm{V}$, and an estimated rated current of $1.31\,\mathrm{A}$. The six slot/eight pole electromagnetic layout avoids a one to one teeth to poles relationship to reduce cogging.

The design process began with motor specification selection, analytical calculations, and SOLIDWORKS modelling. The stator was modelled with a 61 mm outer diameter, six teeth, a central shaft hole, bearing pockets, and a square mounting boss for the test stand. The rotor was modelled as an outrunner can with a 65 mm inner diameter, a 71 mm outer diameter, eight magnet pockets, a shaft clearance hub, and ventilation openings. Both files were exported as STL files and 3D printed. After fabrication, the stator was wound using an ABC winding scheme in a wye connection. The calculated design required approximately 17 turns per tooth; the final physical motor used 8 turns per tooth because slot space and wire gauge constraints prevented achieving the full count.

Testing confirmed that the motor responded to the electronic speed controller and produced slight rotational motion, but the motor did not sustain continuous independent rotation. The main limiting factors were an actual air gap larger than the design target, fewer winding turns than calculated, magnet placement inconsistencies, and mechanical friction from the shaft and bearings. The project completed the full process of designing and building a custom BLDC motor and exposed the practical gap between theoretical motor design and physical electromechanical construction.

Contents

1 Introduction

Brushless DC motors convert electrical energy into mechanical rotation using permanent magnets, stator windings, and electronic commutation. A BLDC motor uses no brushes or mechanical commutator. An electronic speed controller energizes the motor windings by switching motor phases in sequence. A typical three phase BLDC motor achieves improved reliability, lower maintenance, reduced sparking, and better efficiency compared with brushed motor designs.

The objective of this project was to design and build a small BLDC motor that satisfied the course requirements and remained manufacturable using 3D printed parts and available lab components. The motor had to fit within the specified physical size envelope and use the provided shaft, bearings, magnets, copper wire, and test stand.

A radial flux, three phase, six slot/eight pole outrunner BLDC motor was selected. The outrunner layout places the rotor around the stator, increasing the effective rotating radius and improving low speed torque. The six slot/eight pole combination avoids a one to one relationship between stator teeth and rotor poles, which reduces cogging and prevents repeated full alignment between the rotor magnets and stator teeth.

The project followed the engineering workflow below:

Specification selection $\rightarrow$ analytical calculations $\rightarrow$ CAD modelling $\rightarrow$ STL export $\rightarrow$ 3D printing $\rightarrow$ winding $\rightarrow$ assembly $\rightarrow$ ESC testing

The main design goals were to:

- Design a motor that fits within the $74\,\text{mm} \times 74\,\text{mm} \times 44\,\text{mm}$ size constraint.
- Use a six slot/eight pole electromagnetic layout.
- Calculate a realistic number of winding turns per tooth.
- Create manufacturable stator and rotor models in SOLIDWORKS.
- Export STL files suitable for 3D printing.
- Assemble the motor using lab provided shaft, bearings, magnets, and copper wire.
- Test the motor using an electronic speed controller.

2 Technical Background

2.1 BLDC Motor Operation

A brushless DC motor consists of a stationary stator and a rotating rotor. The stator carries wound copper coils and the rotor carries permanent magnets. Current flowing through the stator windings produces magnetic fields that interact with the rotor magnets and generate torque.

Since a BLDC motor contains no brushes, phase switching must be performed electronically. An electronic speed controller energizes selected motor phases in a controlled sequence. In a three phase BLDC motor, two phases carry current at a time and the third phase floats or is monitored depending on the control strategy. This creates a rotating magnetic field that pulls the rotor into motion.

2.2 Outrunner Motor Layout

In an outrunner BLDC motor, the rotor forms an outer rotating shell around the stationary stator. The permanent magnets attach to the inside surface of the rotor and the copper windings sit around the stator teeth. This configuration increases the radius at which magnetic force is applied, improving torque production at low to moderate speeds.

The selected motor uses a radial flux outrunner layout. Magnetic flux crosses the radial air gap between the rotor magnets and stator teeth. The air gap is a critical design parameter because increasing the distance between magnets and windings reduces the magnetic field acting on the stator teeth.

2.3 Slots, Teeth, Poles, and Cogging

The stator teeth provide the locations for winding the copper coils. The rotor poles correspond to the permanent magnets installed around the rotor. The selected design used six stator teeth and eight rotor poles.

The teeth to poles ratio is:

$$T_s : P_r = 6 : 8 \tag{1}$$

This ratio is not one to one because equal stator to rotor alignment increases cogging and makes the motor startup more difficult. The six slot/eight pole layout ensures the teeth and magnets do not repeatedly align in a symmetrical locking position.

3 Design Requirements and Constraints

The motor design was constrained by course requirements and practical manufacturing limits.

3.1 Electrical Requirements

The electrical design targets the following values:

- Rated voltage: $12\,\mathrm{V}$
- Rated torque: $0.075\,\mathrm{N} * \mathrm{m}$
- Rated speed: 2000 RPM
- Estimated current: $1.31\,\mathrm{A}$
- Three phase winding
- Wye connection
- Compatibility with an electronic speed controller

A wye connection reduces the current per phase compared with a delta connection, making the design more suitable for the available wire and course testing setup.

3.2 Mechanical Requirements

The mechanical design satisfies the following requirements:

- Fits within a $74\,\mathrm{mm} \times 74\,\mathrm{mm} \times 44\,\mathrm{mm}$ bounding box.
- Uses 3D printed stator and rotor components.
- Fits the provided shaft and bearing geometry.
- Fits the course test stand.
- Provides clearance between the rotating rotor and stationary stator.
- Provides sufficient slot space for the windings.
- Allows magnet pockets to hold the rotor magnets securely.

3.3 Manufacturing Requirements

The stator and rotor were exported as STL files for 3D printing. The design therefore accounts for printability, tolerances, flat surfaces, support requirements, and clearance for post print assembly.

4 Selected Design Parameters

Table 1 summarizes the main motor parameters used in the design.

Table 1: Selected BLDC motor design parameters.

Parameter	**Value**	**Reason / Note**
Motor type	Radial flux outrunner BLDC	Selected for low speed torque potential
Number of phases	3	Standard BLDC configuration
Stator teeth	6	Multiple of three for three phase winding
Rotor poles / magnets	8	Avoids one to one teeth to poles ratio
Winding scheme	ABC	Simple three phase winding layout
Connection type	Wye / Star	Reduces current per phase
Rated torque	$0.075\,\mathrm{N} * \mathrm{m}$	Target design torque
Rated speed	2000 RPM	Moderate speed target
Rated voltage	$12\,\mathrm{V}$	Within low voltage testing range
Magnet grade	N52 neodymium	High residual flux density
Residual flux density, B_r	$1.42\,\mathrm{T}$	Used for surface flux estimate
Magnet thickness, t	$0.002\,\mathrm{m}$	Matches rotor pocket depth
Air gap, g	$0.002\,\mathrm{m}$ target	Required clearance between rotor and stator
Tooth wall thickness, k	$0.002\,\mathrm{m}$	Plastic wall thickness in flux path
Stator radius, R	$0.0305\,\mathrm{m}$	Used in torque turns calculation
Axial stack length, L	$0.030\,\mathrm{m}$	Selected motor axial length
Calculated turns per tooth	17 turns	From design equation
Actual turns per tooth	8 turns	Limited by slot space and wire thickness

5 Design Calculations

5.1 Angular Velocity

The rated speed was selected as 2000 RPM. Angular velocity was calculated using:

$$\omega = \frac{2\pi N}{60} \tag{2}$$

where N is the speed in RPM. Substituting the selected speed:

$$\omega = \frac{2\pi(2000)}{60} \tag{3}$$

$$\omega = 209.4\,\mathrm{rad/s} \tag{4}$$

5.2 Mechanical Power

Mechanical output power was calculated using:

$$P_m = \tau\omega \tag{5}$$

where τ is torque and ω is angular velocity. At a selected torque of $0.075\,\mathrm{N\,m}$:

$$P_m = (0.075)(209.4) \tag{6}$$

$$P_m = 15.7\,\mathrm{W} \tag{7}$$

5.3 Rated Current

Motor current was estimated from mechanical power and rated voltage:

$$I = \frac{P_m}{V} \tag{8}$$

Substituting the calculated power and selected voltage:

$$I = \frac{15.7}{12} \tag{9}$$

$$I = 1.31\,\mathrm{A} \tag{10}$$

5.4 Flux Density at the Stator Surface

The estimated flux density at the stator surface was calculated using:

$$B_a = B_r \frac{t}{t + g + k} \tag{11}$$

where B_r is the residual flux density of the magnet, t is magnet thickness, g is the air gap, and k is tooth wall thickness. The selected values were:

$$B_r = 1.42\,\mathrm{T}, \quad t = 0.002\,\mathrm{m}, \quad g = 0.002\,\mathrm{m}, \quad k = 0.002\,\mathrm{m} \tag{12}$$

Therefore:

$$B_a = 1.42 \left(\frac{0.002}{0.002 + 0.002 + 0.002} \right) \tag{13}$$

$$B_a = 0.473\,\mathrm{T} \tag{14}$$

This equation shows the importance of air gap control. An actual air gap larger than the design value reduces the flux density at the stator surface and lowers the torque the motor can produce.

5.5 Number of Teeth per Phase

The number of teeth per phase was calculated using:

$$n = \frac{\text{Total number of stator teeth}}{\text{Number of phases}} \tag{15}$$

For six teeth and three phases:

$$n = \frac{6}{3} = 2 \tag{16}$$

5.6 Turns per Tooth

The required number of turns per tooth was estimated using:

$$N = \frac{\tau}{4nB_a LRI} \tag{17}$$

where N is the turns per tooth, τ is torque, n is teeth per phase, B_a is flux density at the stator surface, L is axial stack length, R is stator radius, and I is current.

Substituting the selected values:

$$N = \frac{0.075}{4(2)(0.473)(0.030)(0.0305)(1.31)} \tag{18}$$

$$N = 16.5 \approx 17 \text{ turns per tooth} \tag{19}$$

The calculated value falls within the practical target range of 5 to 50 turns per tooth. The final physical motor used 8 turns per tooth because the available wire gauge and stator slot volume could not accommodate the full calculated number.

6 CAD Design

The motor was modelled in SOLIDWORKS using separate stator and rotor parts. The CAD design covers electromagnetic geometry, shaft alignment, bearing support, magnet pockets, and manufacturability through 3D printing.

6.1 Stator Design

The stator was designed first. A circle with a radius of 30.5 mm defines the outer radius of the stator teeth, giving a total stator diameter of 61 mm. A smaller concentric circle with a radius of 20 mm defines the inner hub and back iron region.

A single tooth profile was created with radial geometry and a tooth width of approximately 6 mm. A circular pattern with six instances was used to complete the tooth array. The stator was extruded to an axial length of 30 mm.

A square mounting boss of 15 mm $\times$ 15 mm was added to the bottom face of the stator. This boss fits the stator into the course test stand and prevents the stator from rotating during motor operation.

A central shaft hole was cut with a radius of 1.5 mm for the 3 mm shaft. Bearing pockets were cut into the stator to locate the bearings and improve shaft alignment.

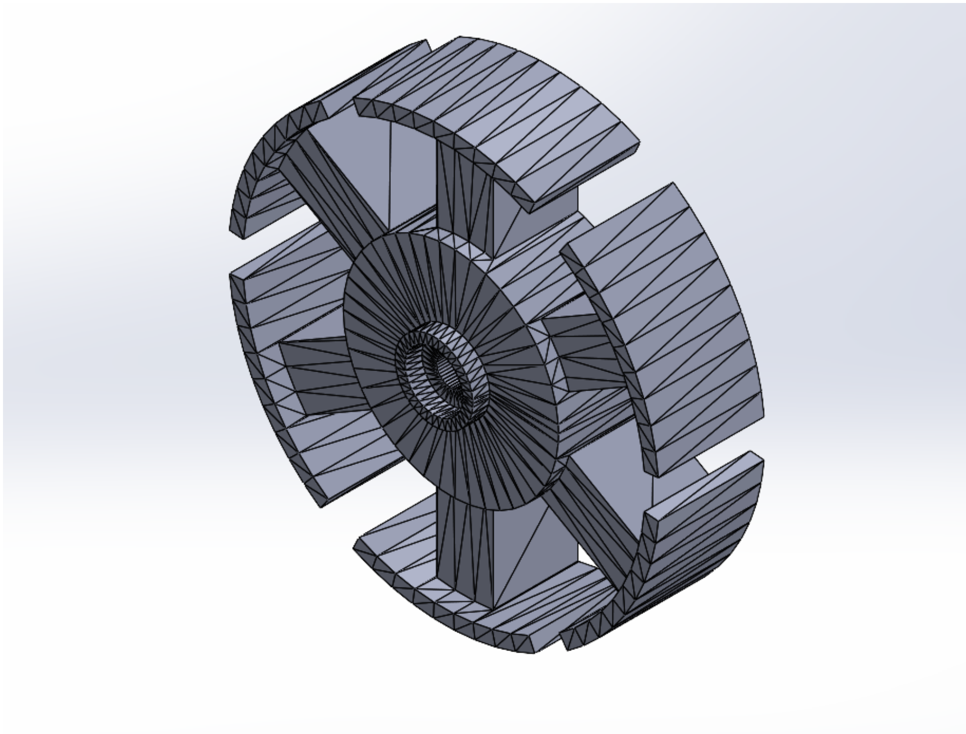

Figure 1: Isometric view of the six slot outrunner stator CAD model

6.2 Rotor Design

The rotor was designed as an outrunner can. To maintain a target air gap of 2 mm, the inside radius of the rotor was set as:

$$R_{\text{rotor, inner}} = R_{\text{stator}} + g = 30.5 + 2.0 = 32.5\,\text{mm} \tag{20}$$

This gives an inner diameter of 65 mm. The outer radius of the rotor was set to 35.5 mm, giving an outer diameter of 71 mm. The annular rotor wall was extruded to an axial length of 30 mm.

A 3 mm thick cap closes the top of the rotor. Hub cylinders near the shaft region improve strength at the shaft and bearing interface. A shaft clearance hole with radius 1.65 mm passes through the rotor hub to allow a sliding fit.

Eight rectangular magnet pockets were cut into the inside surface of the rotor and patterned around the rotor to create the eight pole layout. The magnet pocket depth matches the magnet thickness of approximately 2 mm. Ventilation slots in the rotor cap reduce mass and allow air flow near the windings.

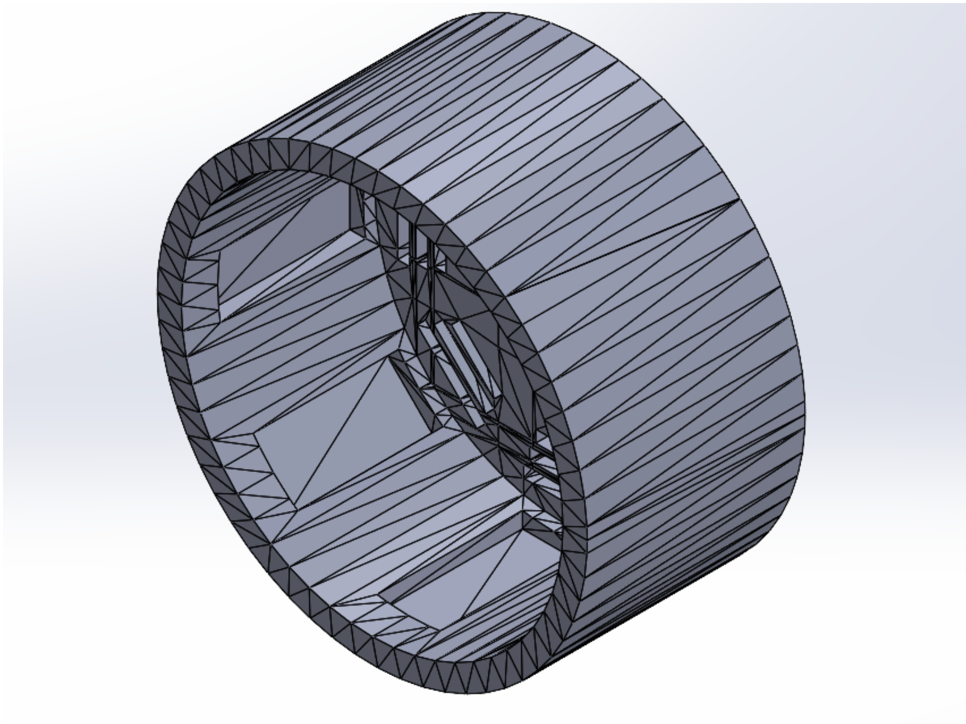

Figure 2: Rotor CAD model with eight internal magnet pockets

6.3 SOLIDWORKS Assembly

After modelling the individual parts, a full assembly was created in SOLIDWORKS. The stator bottom, stator top, and rotor were inserted into a SOLIDWORKS assembly file. The stator components were aligned to form the complete stationary structure and the rotor was positioned concentrically around the stator to represent the final outrunner motor geometry.

The assembly confirmed the relative position of the rotor and stator and produced documentation screenshots for the final report. Because the available geometry was based on STL mesh files, the assembly served as a visual and documentation model rather than a fully parametric SOLIDWORKS assembly.

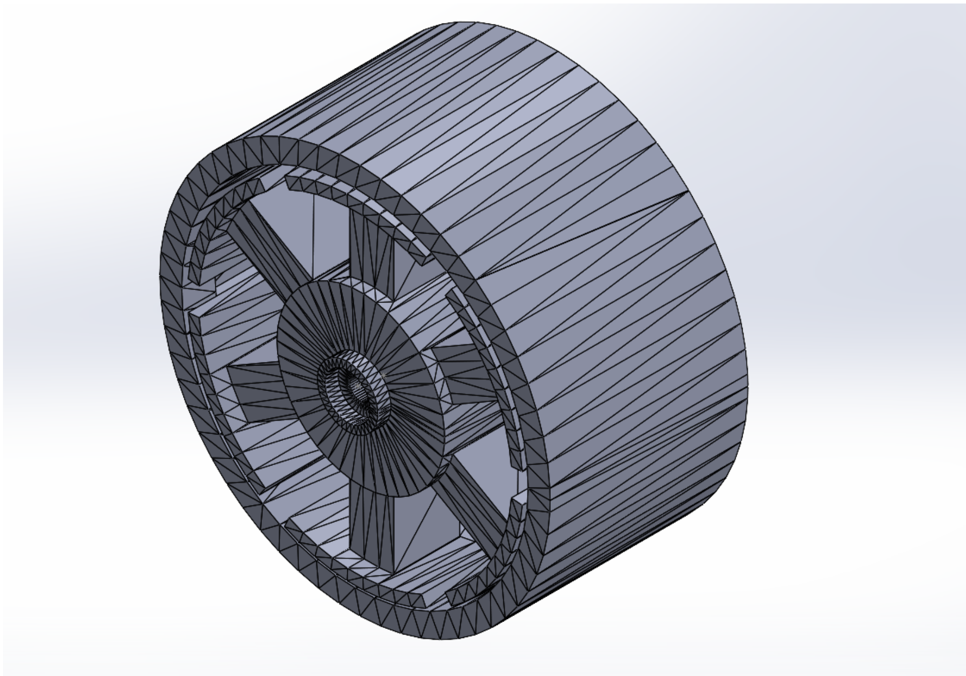

Figure 3: Full SOLIDWORKS assembly of the six slot/eight pole outrunner BLDC motor

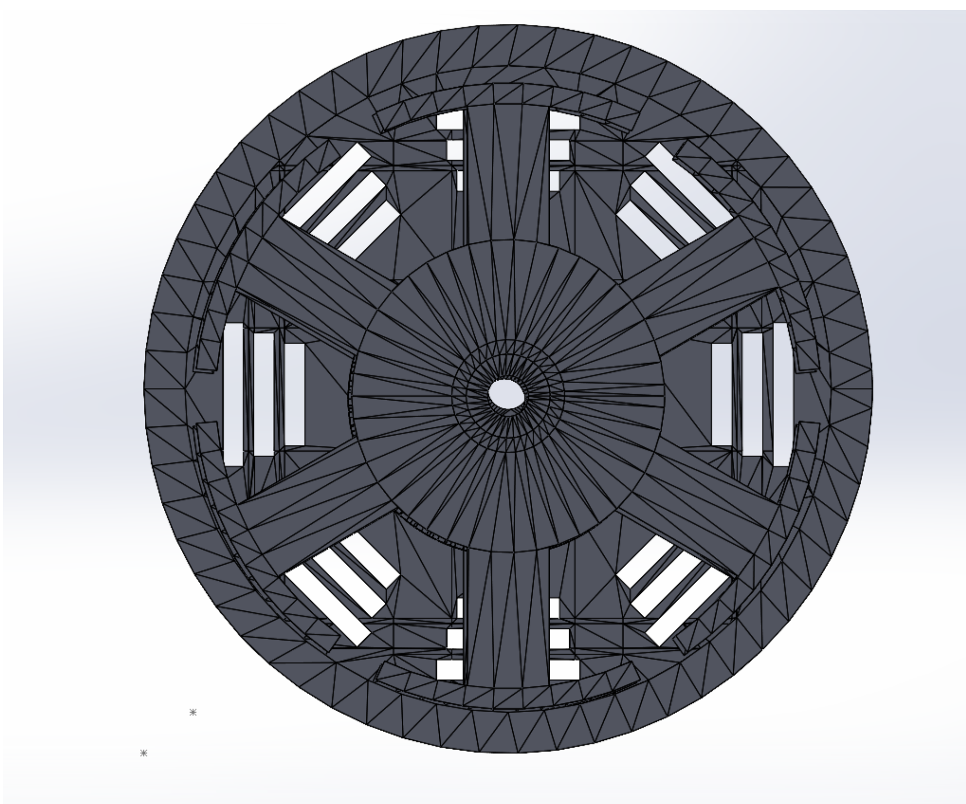

Figure 4: Front view of the assembled BLDC motor showing the rotor can aligned around the stator

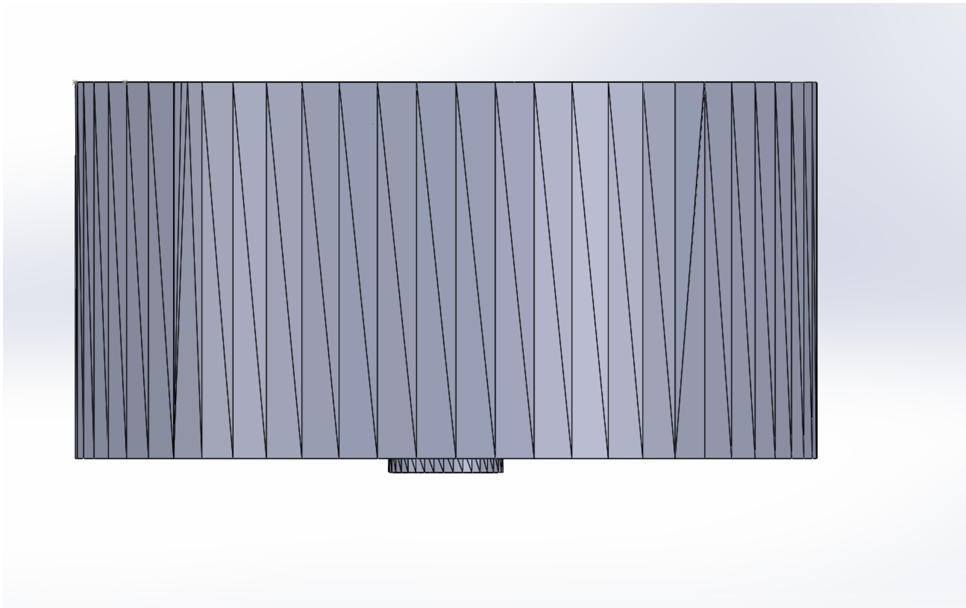

Figure 5: Top view of the rotor and stator alignment in the BLDC motor assembly

7 3D Printing and Fabrication

After completing the CAD models, the stator and rotor were exported as STL files for 3D printing. The exported files were:

- `Rotor.STL`
- `StatorBottom.STL`
- `StatorTop.STL`

The stator was split into top and bottom printed sections to improve manufacturability and simplify the assembly around the bearing and shaft features. The rotor was printed as an outrunner can with integrated magnet pockets and ventilation slots.

After printing, the parts were inspected for fit and clearance. The shaft was installed through the central hole and the rotor was placed around the stator. The printed assembly fit within the size constraint, but the actual air gap exceeded the target design gap because of 3D printing tolerance and the need to prevent rubbing between the rotating and stationary parts.

Figure 6: Assembled 3D printed BLDC motor with shaft installed

8 Winding and Motor Assembly

8.1 Winding Scheme

The selected winding scheme uses an ABC three phase arrangement with a wye connection. The wye connection reduces the phase current compared with a delta connection, making it more suitable for the available wire and low current test setup.

The calculated design required approximately 17 turns per tooth. The final motor used 8 turns per tooth because the available lab wire was too thick to fit the full calculated turn count inside the stator slots. All windings were wound clockwise per the selected winding direction.

8.2 Magnet Installation

Eight N52 neodymium magnets were inserted into the rotor pockets. The magnets were arranged around the inside of the rotor to form the eight pole magnetic field. Correct magnet placement is important because uneven spacing or incorrect polarity creates unbalanced magnetic pull and reduces startup performance.

8.3 Physical Assembly

The printed stator was installed onto the test stand using the square mounting boss. The shaft and bearings were installed through the central axis and the rotor was mounted concentrically around the stator. The final assembly includes the wound stator, rotor, shaft, bearings, magnets, and base.

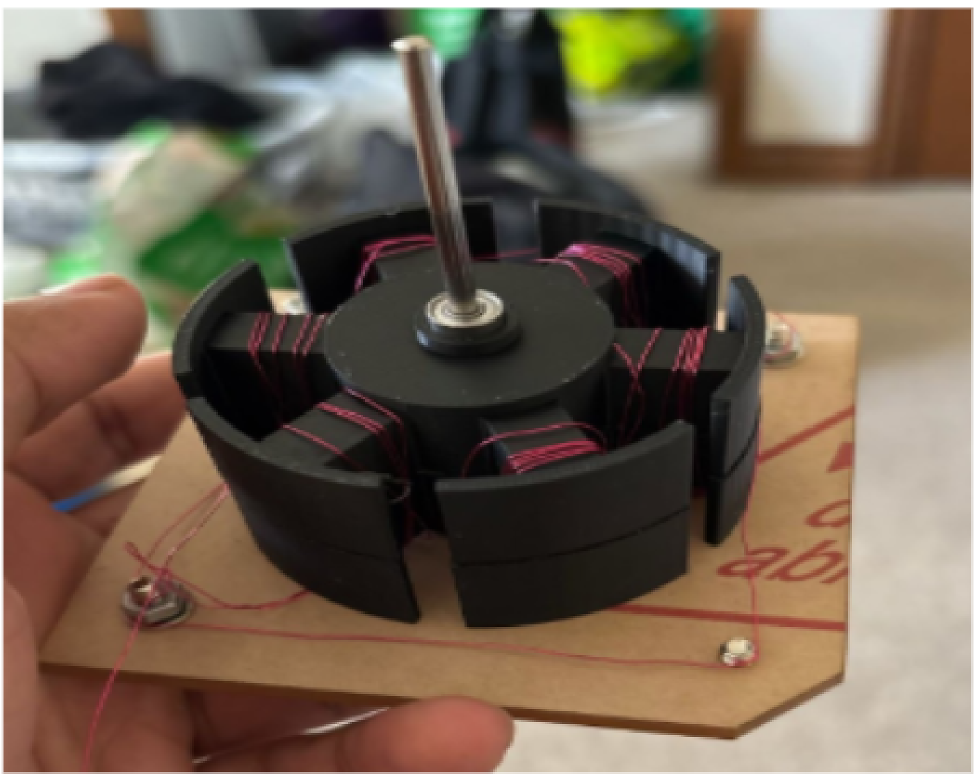

Figure 7: Stator assembly with wound coils installed on the test base

9 Testing Methodology

The motor was tested using an electronic speed controller, power supply, and control input. The ESC was connected to the three motor phase wires and attempted to start the motor using electronic commutation.

Before testing, the following checks were performed:

- The rotor was checked for free movement around the stator.
- The shaft and bearings were checked for alignment.
- The magnet pockets were inspected for magnet placement.
- The windings were inspected for loose wires or obvious shorts.
- The phase wires were connected to the ESC.
- The supply voltage was kept within the course voltage limit.

During testing, the motor response was observed. The main evaluation criteria were whether the motor could start, whether it could sustain rotation, whether the rotor contacted the stator, and whether the ESC detected and drove the motor phases correctly.

10 Results

Table 2 summarizes the final design and testing results.

Table 2: Final BLDC motor parameter summary.

Parameter	**Value**	**Notes**
Motor type	Radial flux outrunner	Three phase BLDC motor
Stator teeth	6	Compatible with three phase winding
Rotor poles	8	Avoids $T : P = 1$
Winding scheme	ABC, wye	Selected to reduce phase current
Magnet grade	N52 neodymium	$B_r = 1.42\,\mathrm{T}$
Rated torque	$0.075\,\mathrm{N} * \mathrm{m}$	Design target
Rated speed	2000 RPM	Moderate speed target
Rated voltage	12 V	Within voltage constraint
Mechanical power	15.7 W	From $P_m = \tau\omega$
Rated current	1.31 A	From $I = P/V$
Target air gap	0.002 m	Actual gap became larger after printing
Stator radius	0.0305 m	Used in turn calculation
Axial length	0.030 m	3D printable stack length
Flux density at stator	0.473 T	From flux density equation
Calculated turns per tooth	17 turns	From design equation
Actual turns per tooth	8 turns	Limited by wire thickness and slot space
Magnet thickness	0.002 m	Matches rotor pocket depth
Size constraint	74 mm × 74 mm × 44 mm	Satisfied
Testing outcome	Partial rotation	Motor did not sustain continuous operation

The final motor fit within the required size envelope and was assembled using the 3D printed rotor and stator components. The motor responded when connected to the ESC and produced slight rotational movement but did not sustain continuous independent rotation.

The design succeeded in completing the CAD model, exporting STL files, 3D printing, assembling with the shaft and bearings, and implementing the three-phase winding. The main performance limitations were the larger than target air gap, reduced actual turns per tooth, magnet placement variation, and friction in the shaft-bearing assembly.

11 Discussion

The final motor performance did not match the theoretical design calculations. The motor produced slight motion when the ESC attempted to start it but could not maintain continuous rotation. Several factors account for this outcome.

The first issue was the air gap between the rotor magnets and the stator teeth. The design target was approximately 2 mm. After 3D printing and physical assembly, the actual air gap exceeded the target value. The flux density equation depends on air gap thickness, so any increase in air gap reduces the magnetic field acting on the stator teeth. A weaker magnetic field produces less electromagnetic torque, making startup and sustained rotation more difficult.

The second issue was the reduced winding count. The calculation predicted approximately 17 turns per tooth, but only 8 turns per tooth could be wound using the available wire. The magnetic field produced by a coil depends on the number of turns and current, so reducing the turns reduced the available stator field strength. This lowered the torque and reduced the back EMF available to the ESC, making sensorless startup more difficult.

The third issue was magnet placement. The rotor pockets held the magnets, but small inconsistencies in spacing or alignment created uneven magnetic pull. This makes the rotor harder to start and increases vibration and cogging.

The fourth issue was mechanical friction. Even though the CAD assembly showed no interference, the printed parts, shaft, and bearings introduced real world tolerances. Slight misalignment or bearing resistance can consume a large portion of the available starting torque, particularly in a small motor with reduced winding turns.

The design approach was solid. The six slot/eight-pole layout was appropriate, the wye connected ABC winding was reasonable, and the motor fit within the size envelope. The gap between the calculation and the final result came from physical implementation constraints rather than from the overall concept.

12 Limitations

The project had several limitations.

First, the stator and rotor were 3D printed in plastic rather than manufactured with laminated ferromagnetic steel. A commercial BLDC motor uses magnetic steel laminations to guide flux through the stator. The plastic stator structure reduced the magnetic effectiveness of the design.

Second, the actual air gap was larger than the target value. A larger air gap reduced the magnetic coupling between the rotor magnets and stator windings and decreased torque production.

Third, the number of turns per tooth was lower than calculated. The calculated design required 17 turns per tooth, but the final motor used only 8 turns per tooth because of wire thickness and slot space limitations.

Fourth, testing was observational. The project did not include quantitative measurements of torque, no load speed, phase current, back EMF, or efficiency. Performance could only be evaluated based on whether the motor started and sustained rotation.

Fifth, magnet placement and winding were performed by hand. Manual assembly introduced small inconsistencies that affected the final motor behaviour.

13 Future Work

Future improvements should focus on increasing torque production and reducing mechanical losses.

The first improvement is reducing the air gap. A smaller, more consistent air gap increases the magnetic flux density at the stator surface and improves torque. Better print accuracy, a redesigned rotor stator clearance, and improved alignment features would achieve this.

The second improvement is increasing the available winding space. The stator slot geometry can be redesigned to accommodate more copper turns. Thinner magnet wire would allow the calculated 17 turns per tooth to be achieved.

The third improvement is tightening magnet alignment. Tighter rotor pockets, adhesive retention, and polarity markings would reduce magnet placement errors and improve rotor balance.

The fourth improvement is reducing mechanical friction. Better bearing alignment, a smoother shaft fit, and improved hub geometry would reduce startup resistance.

The fifth improvement is adding quantitative testing. Future testing should measure phase resistance, phase current, no load speed, startup current, back EMF, and torque. These measurements would allow the physical motor performance to be compared with the design calculations.

14 Conclusion

This project completed the full process of designing and building a custom brushless DC motor. A radial flux, three phase, six slot/eight pole outrunner BLDC motor was designed through analytical calculations, SOLIDWORKS CAD modelling, STL export, 3D printing, manual winding, assembly, and ESC based testing.

The design satisfied the main physical constraints and produced a manufacturable motor assembly. The stator and rotor were modelled, printed, assembled with the shaft and bearings, wound using an ABC wye configuration, and connected to an electronic speed controller.

The final motor produced slight rotational motion but did not sustain continuous independent operation. The main reasons were the larger than target air gap, the reduced actual winding count, magnet placement variation, and mechanical friction. These issues reduced the available torque and made ESC startup more difficult.

The project demonstrated that correct calculations alone do not guarantee physical performance. Motor performance depends on manufacturing tolerance, winding feasibility, air gap control, magnet placement, and mechanical alignment. The project was successful as an engineering design and learning exercise because it connected theory, CAD, fabrication, winding, assembly, and practical testing into one complete electromechanical system.

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